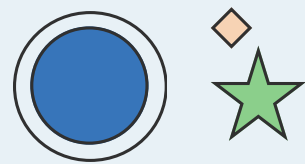


Controlled visual tasks

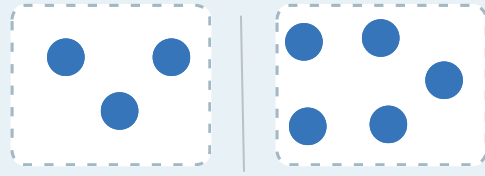
Visual primitives

Feature binding



Color of the outlined object?

Numerosity



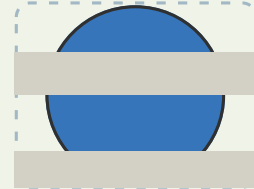
Which side has more?

Spatial relations



Circle relative to star?

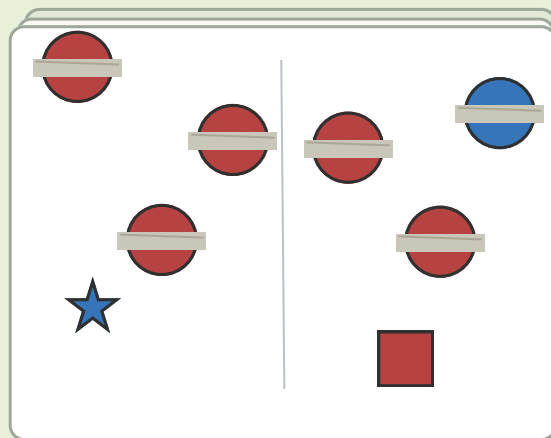
Amodal completion



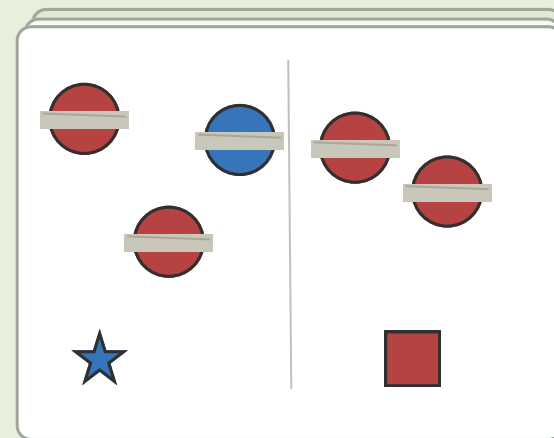
Hidden shape?

Composite + counterfactual pairs

Which side contains more red circles?



Answer: left

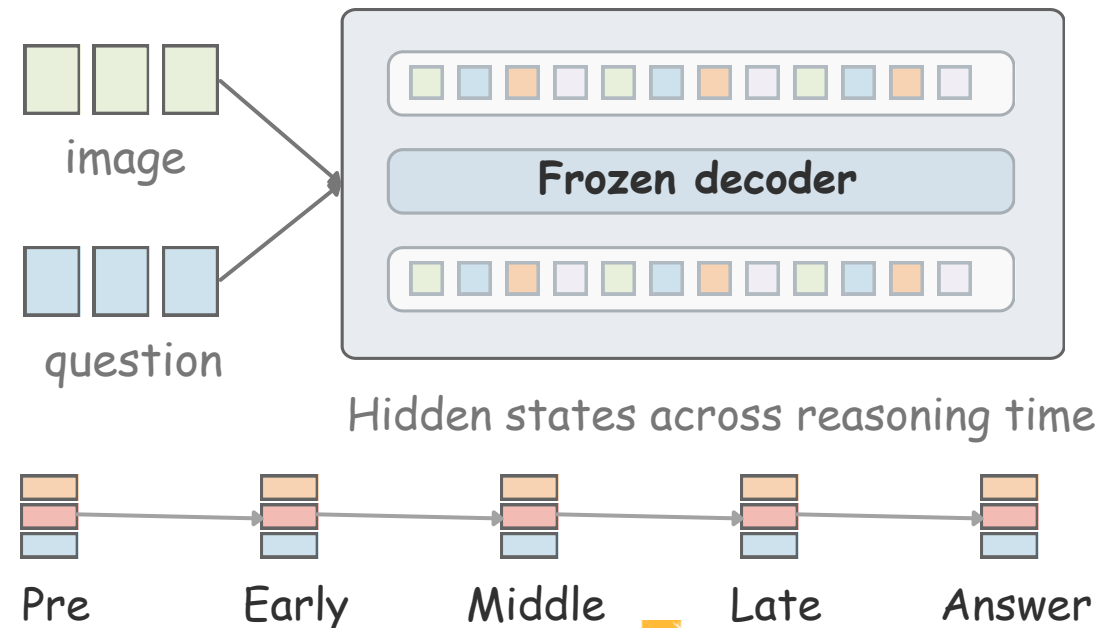


Answer: right

Target colors exchanged

Inside native reasoning

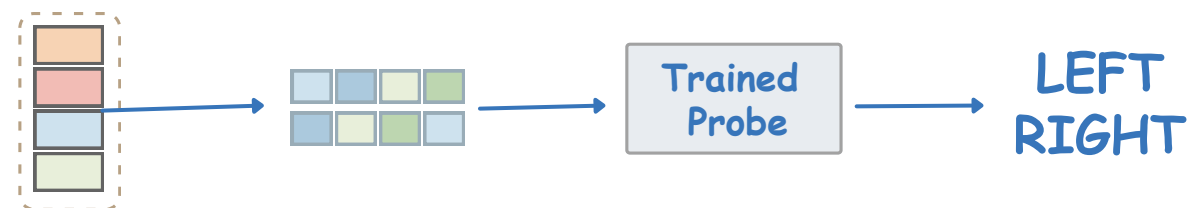
VLM



Answer-readiness analysis

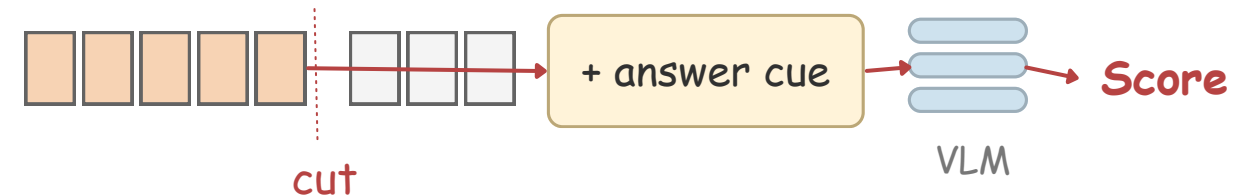
Linear probing

Can the answer be linearly decoded from the state?



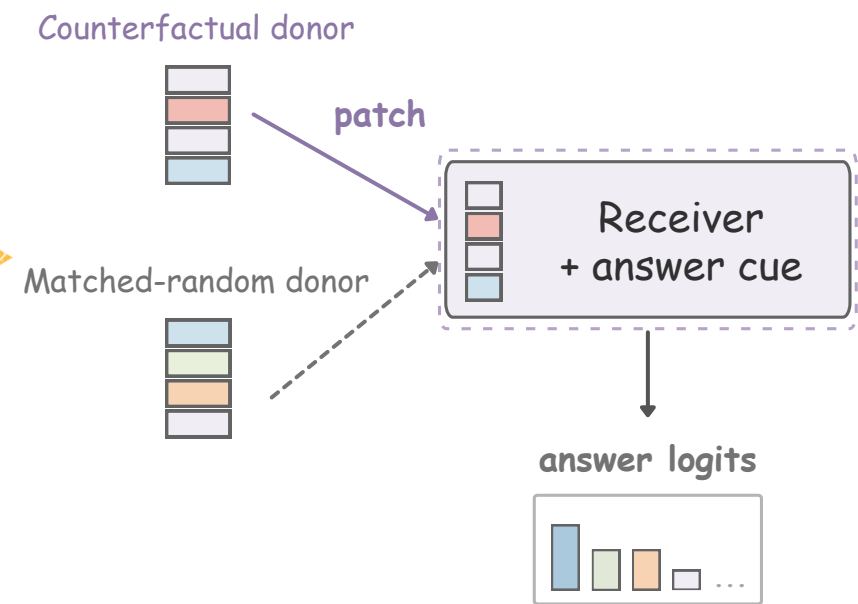
Prefix answerability

Can the model answer at this checkpoint?



Causal influence

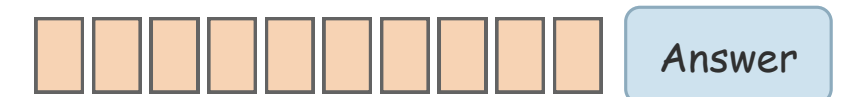
Does the primitive state affect the answer?



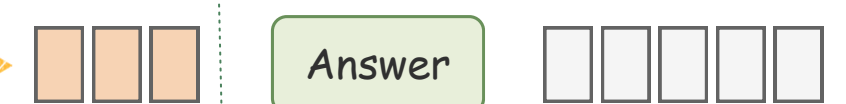
Compare donor-relative margin

Adaptive stopping

Full thinking



Adaptive



stop

State-based readiness detector

max 86.9% fewer reasoning
max +15.2 pp accuracy